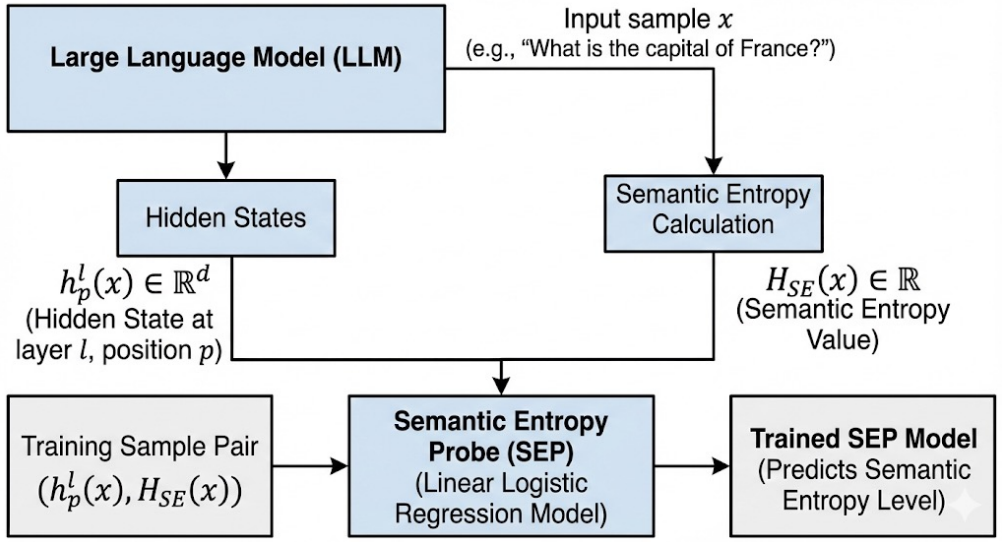
SEPs

SEPs[50]

Training SEPs

Semantic Entropy Probes (SEPs) are linear logistic regression models trained on LLM hidden states to predict semantic entropy.

Each training sample pairs a hidden representation $h_p^l(x) \in \mathbb{R}^d$ with its corresponding semantic entropy $H_{SE}(x) \in \mathbb{R}$ for input x.



Binarization

The semantic entropy $H_{SE}(x)$ is a continuous value and is binarized to enable classification into high- and low-entropy groups:

$$\tilde{H}_{SE}(x) = \mathbf{1}\left[H_{SE}(x) > \gamma^*\right].$$

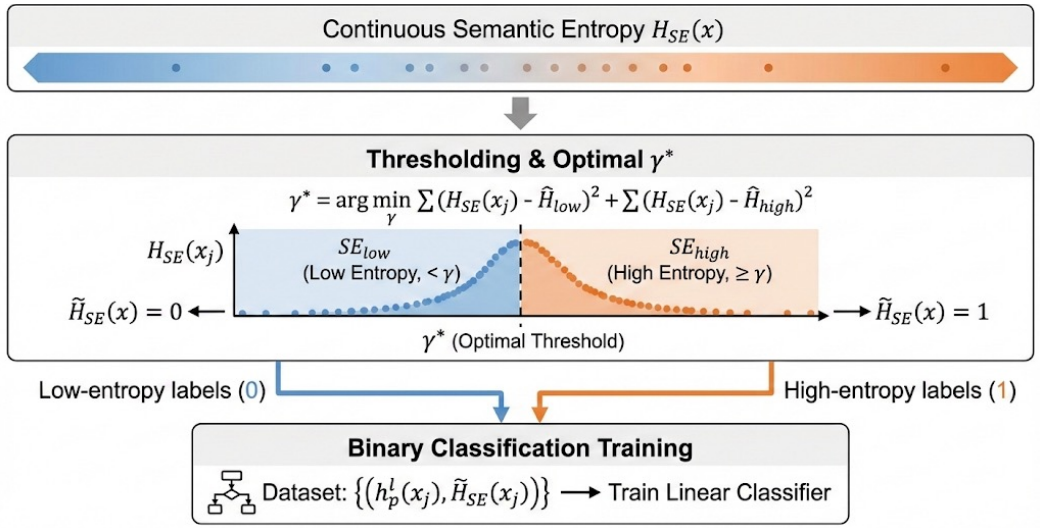
The threshold is chosen to minimize within-group variance:

$$\gamma^* = \arg\min_{\gamma} \sum_{j \in SE_{\text{low}}} \left(H_{SE}(x_j) - \hat{H}_{\text{low}}\right)^2 + \sum_{j \in SE_{\text{high}}} \left(H_{SE}(x_j) - \hat{H}_{\text{high}}\right)^2,$$

where

$$SE_{\text{low}} = \{j : H_{SE}(x_j) < \gamma\}, \qquad SE_{\text{high}} = \{j : H_{SE}(x_j) \geq \gamma\}.$$

The resulting dataset $\left\{\left(h_p^l, (x_j)\tilde{H}_{SE}(x_j)\right)\right\}$ is used to train a linear classifier for predicting high semantic entropy.



[50] Semantic Entropy Probes: Robust and Cheap Hallucination Detection in LLMs, arxiv 2024

SEPs[50]

- **Task Type:** Open-ended generation

- **Introduction:** Semantic Entropy Probes (SEPs) train a lightweight linear model on LLM hidden states to approximate semantic entropy. The probe is trained with hidden-state/semantic-entropy pairs supervision and outputs uncertainty estimates via a single forward pass at inference. SEPs provides an efficient and scalable semantic-level uncertainty estimator while avoiding the computational cost of sampling-based entropy measures.

Type	Uncertainty Sources	Model Accessibility	Training Necessity	Consumption	Evaluation
Single-pass	Decoding	White-box	Fine-tuning	High	Ranking

[50] Semantic Entropy Probes: Robust and Cheap Hallucination Detection in LLMs, arxiv 2024